BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

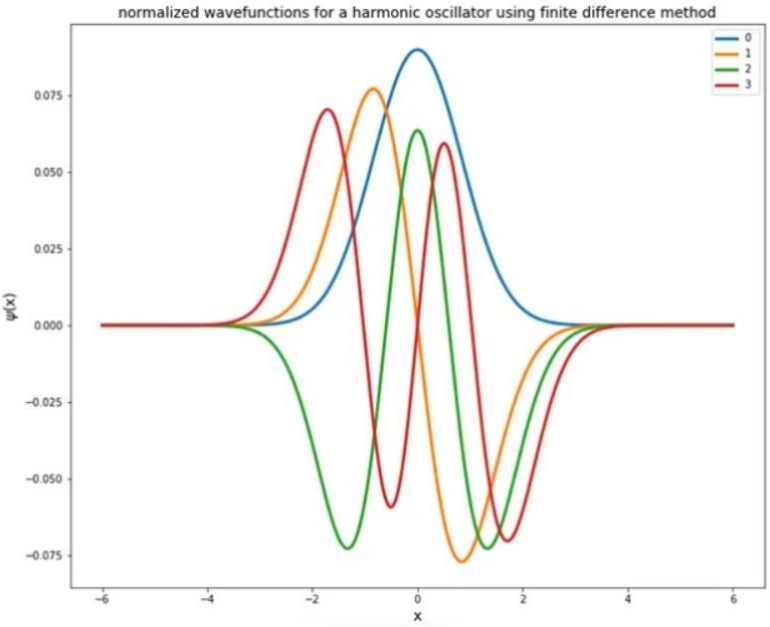
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Course No. : EEE 212

Course Name : Numerical Analysis Laboratory

Name of Project :

SOLVING TIME INDEPENDENT SCHRODINGER'S EQUATION



Submitted To:

1. *Sadman Sakib Ahbab*
   Lecturer
   Dept. of EEE, BUET
2. *Dibakar Roy*
   Graduate Teaching Assistant
   Dept. of EEE, BUET

Submitted By:

Team Name : **MatLite**

Team Member 1:
Name : *Ilias Ahmad*
ID : ***2006068***
Section : B1

Team Member 2:
Name : *Nahian Tasnim*
ID : ***2006098***
Section : B1

Date of Submission : 06 March 2023

Objective:

- To calculate the Eigen state energy for a particle confined in finite potential well is the main objective of this project.
- To compare between finite and infinite potential well is also a desired outcome from this project.

Theory:

The time-independent Schrodinger equation is a linear ordinary differential equation that describes the wavefunction or state function of a quantum-mechanical system. The solution of the Schrodinger equation yields quantized energy levels as well as wavefunctions of a given quantum system.

In quantum mechanics, the particle in a box model (also known as the infinite potential well or the infinite square well) describes a particle free to move in a small space surrounded by impenetrable barriers. The model is mainly used as a hypothetical example to illustrate the differences between classical and quantum systems.

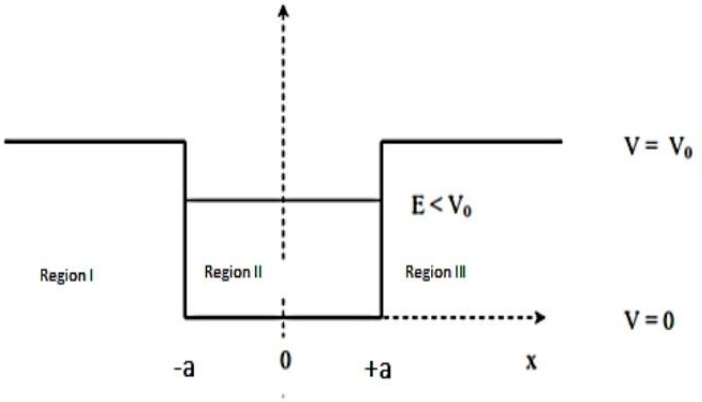


Fig.(1): A finite square well, depth, V_o , width 2a .

The finite potential well (also known as the finite square well) is an extension of the infinite potential well, in which a particle is confined to a "box", but one which has finite potential "walls". Unlike the infinite potential well, there is a probability associated with the particle being found outside the box. The quantum mechanical interpretation is unlike the classical interpretation, where if the total energy of the particle is less than the potential energy barrier of the walls it cannot be found outside the box. In the quantum interpretation, there is a non-zero probability of the particle being outside the box even when the energy of the particle is less than the potential energy barrier of the walls.

For the one-dimensional case on the x-axis, the Time-Independent Schrodinger's equation can be written as:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_{(x)})\psi = 0$$

where,

$\hbar = \frac{h}{2\pi}$ is the reduced Plank's constant

m = mass of the particle

ψ = the wavefunction that we want to find

$V_{(x)}$ = function describing the potential energy at each point x

E = eigen energy

We use finite difference method to solve this equation. In finite difference method, x is divided into a grid of equally spaced points (should be as small as possible). We will use the properties of 'central difference method'.

$$\left.\frac{d\psi(x)}{dx}\right|_{x=x_i} \approx \left.\frac{\Delta\psi(x)}{\Delta x}\right|_{x=xi} = \frac{(\Psi_{x_{i+1}} - \Psi_{x_{i-1}})}{(x_{i+1} - x_{i-1})} = \frac{(\Psi_{i+1} - \Psi_{i-1})}{2d}$$

$$\left.\frac{d^2\psi(x)}{dx^2}\right|_{x=x_i} = \frac{(\Psi_{i+1} - 2\Psi_i + \Psi_{i-1})}{d^2} = \frac{2m}{\hbar^2}\left(V_{o(x)} - E\right)\psi(x)$$

Or,

$$-\frac{\hbar^2}{2md^2}(\Psi_{i+1} - 2\Psi_i + \Psi_{i-1}) + v_i\psi_i = E\psi_i$$

Now , we set up our own system. At each point (1,2,3,…), we have value for wave function ψ_i , and external potential V_i . We get various wavefunction for different energy states of an electron for given external potential.

Here, we write matrices equation for solving:

$$\left(-\frac{\hbar^2}{2md^2}\right)\begin{bmatrix} -2 & 1 & 0 & . & . & . \\ 1 & -2 & 1 & 0 & . & . \\ 0 & 1 & -2 & 1 & 0 & . \\ . & 0 & . & . & . & . \\ . & . & . & . & . & 1 \\ . & . & . & . & 1 & -2 \end{bmatrix}\begin{bmatrix} \psi 1 \\ \psi 2 \\ . \\ . \\ . \\ \psi n \end{bmatrix} + \begin{bmatrix} v1 & 0 & . & . & . & 0 \\ 0 & v2 & & & & . \\ 0 & & v3 & & & . \\ . & & & . & & . \\ . & & & & . & . \\ 0 & . & . & . & . & vn \end{bmatrix}\begin{bmatrix} \psi 1 \\ \psi 2 \\ . \\ . \\ . \\ \psi n \end{bmatrix} = E\begin{bmatrix} \psi 1 \\ \psi 2 \\ . \\ . \\ . \\ \psi n \end{bmatrix}$$

Thus,

$K\psi + V\psi = E\psi$

Or, $(K + V)\,\psi = E\psi$

Or, $H\psi = E\psi$

Where, H = Hamiltonian Matrix

E = Eigen-Energy Matrix

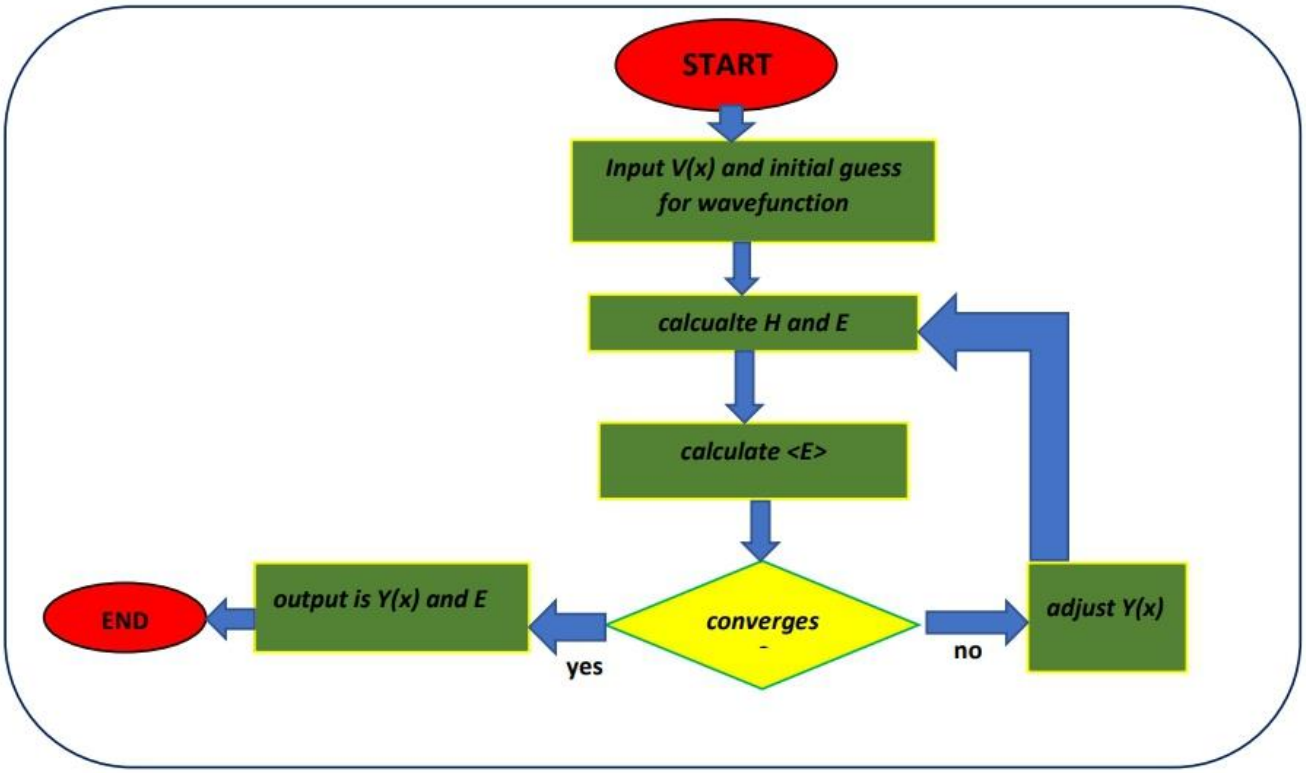
From our given system, we can design H, from where we can solve for E, ψ .

Now, eigen value for H = $[\psi_i \quad E]$, which is a matrix containing wavefunctions for different energy states.

Algorithm:

- **Step 1**: Start
- **Step 2**: Input the value of *V(x)* and initial guess for wavefunction *ψ(x).*
- **Step 3**: Calculate Hamiltonian matrix H and eigen vector E.
- **Step 4**: Calculate <E>.
- **Step 5**: Do the values converge?
    - If no, go back to step 3
    - If yes, output is *ψ(x)* and E.
- **Step 6**: End

Flowchart:



Code:

```
clc;
clear ;
close all;
%%
disp("    ================================================================================")
disp("                                    Team MATLITE  ")
disp("    ================================================================================")
%% Define units
global m nm kg sec J eV mass_e h_bar well_width check
m = 1;
kg = 1;
sec = 1;
J = 1;
nm = 1E-9 *(m);
%%

eV = 1.60218E-19 *(J);

mass_e = 9.10938E-31 *(kg);
h_bar = (6.62607E-34)/(2*pi) *(J*sec);



%% Taking inputs

fprintf('    Potential Well Type: ')
fprintf( '\n\n        i.Input 1 for Finite \n       ii.Input 2 for Ininite\n\n');
fprintf('       >>');
num = input('');

%%
well_width=0;
high_p=0;
x_rag=0;
dmin=0;

check = 0;

while(is_inside(1*nm,20*nm,well_width))
if(check) disp('Invalid input')
end
fprintf('\r    Potential Well Width(between 1-20 (nm)): ');
well_width = input('')*(nm);
check = is_inside(1*nm,20*nm,well_width);
end

check = 0;

while(is_inside(0.01*eV,1*eV,high_p))
if(check) disp('Invalid input')
```

```
end
fprintf('\r    Higher potential (between 0.01-1 (eV)): ');
high_p = input('')*(eV);
check = is_inside(0.01*eV,1*eV,high_p);
end

check = 0;

while(is_inside(10*nm,50*nm,x_rag))
if(check) disp('Invalid input')
end
fprintf('\r    Full potential well length  (between 10-50 (nm)): ')
x_rag = input('')*(nm);
check = is_inside(10*nm,50*nm,x_rag);
end

check = 0;

while(is_inside(0.001*nm,0.1*nm,dmin))
if(check) disp('Invalid input')
end
fprintf('\r    Lowest difference (between 0.001-0.1 (nm)): ')
dmin = input('')*(nm);
check = is_inside(0.001*nm,0.1*nm,dmin);
end

%% For Finite Well
if num ==1

%% Determinate simulation parameters
dx = dmin;
xRange = x_rag;
qWell_wide = well_width;
V_upper = high_p;
V_lower = 0.00 *(eV);

%% Preallocating variables and functions
x = (-xRange/2:dx:xRange/2);

%% constructure the quantum well
V = zeros(size(x));

V(x<-qWell_wide/2) = V_upper;
V(x>qWell_wide/2) = V_upper;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

V(x>=0 & x<=qWell_wide/2) = V_lower;
V(x<=0 & x>=-qWell_wide/2) = V_lower;
```

```
figure(1), plot(x/nm, V/eV,'LineWidth',5)
xlim([min(x/nm),max(x/nm)]), ylim([min(V/eV),max(1.4*V/eV)]),grid on
 title('FiniteWell')
xlabel('x [nm]'),ylabel('V (eV)')

%% construct the Hamiltonian operator
H0 = diag((h_bar^2/(mass_e*dx^2)) + V);
H1 = diag(ones(1,length(x)-1)*(-h_bar^2/(2*mass_e*dx^2)),1);
H2 = diag(ones(1,length(x)-1)*(-h_bar^2/(2*mass_e*dx^2)),-1);
H = H0+H1+H2;

%% Calculate field propagation
[psi,En] = eig(H);

%%
plotModes = sum(diag(En)>V_lower & diag(En)<V_upper);
figure(1)
for plotMode = 1:plotModes
    rescale = ((En(1,1)-V_lower))/(max(psi(:,plotMode))-min(psi(:,plotMode)));
    psi_rescal_plot = psi(:,plotMode)*rescale + En(plotMode,plotMode);
    hold on, plot(x/nm, psi_rescal_plot/eV,'LineWidth',3)
    hold on, plot(x/nm, En(plotMode, plotMode)*ones(size(x))/eV,'LineWidth',1.2)

end


end


%% For Infinite Well

if num ==2
 %% Determinate simulation parameters
dx = dmin;
xRange = x_rag;
qWell_wide = well_width;
V_upper = high_p;
V_lower = 0.00 *(eV);

%% Preallocating variables and functions

x = (-xRange/4:dx:xRange/4);
y = (-xRange:dx:xRange);

%% constructure the quantum well
V = zeros(size(x));

V(y<-qWell_wide/2) = V_upper;
V(y>qWell_wide/2) = V_upper;
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

V(y>0 & y<qWell_wide) = V_lower;
V(y<0 & y>-qWell_wide) = V_lower;

figure(1), plot(y/nm, V/eV,'LineWidth',4)
xlim([min(y/nm),max(y/nm)]), ylim([min(V/eV),max(1.1*V/eV)]),grid on
title('InfiniteWell')
xlabel('x [nm]'),ylabel('V (eV)')

%% construct the Hamiltonian operator
H0 = eye(length(x),length(x))*(-2)*(-h_bar^2/(2*mass_e*dx^2));
H1 = diag(ones(1,length(x)-1)*(-h_bar^2/(2*mass_e*dx^2)),1);
H2 = diag(ones(1,length(x)-1)*(-h_bar^2/(2*mass_e*dx^2)),-1);
H = H0+H1+H2;

%% Calculate field propagation
[psi,En] = eig(H);

%%
plotModes = sum(diag(En)>V_lower & diag(En)<V_upper);
figure(1)
for plotMode = 1:plotModes
    rescale = ((En(1,1)-V_lower))/(max(psi(:,plotMode))-min(psi(:,plotMode)));
    psi_rescal_plot = psi(:,plotMode)*rescale + En(plotMode,plotMode);
    E = En(plotMode, plotMode)*ones(size(x));
    hold on, plot(x/nm, psi_rescal_plot/eV,'LineWidth',3)
    hold on, plot(x/nm, E/eV,'LineWidth',1.2)
end
end

%%
disp('')
disp('    Team Members-')
disp('')
disp('        1.Ilias Ahamd (2006068)')
disp('        2.Nahian Tasnim (2006098)');

disp('')
disp('    Thank you')

%% checking Function
function y = is_inside(a,b,c)
  if( a<=c && b>=c ) y = 0;

else y = 1;
end
end
```

Test Condition:

Test conditon 1: (Finite Well)

```
Command Window
================================================================================
                                Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

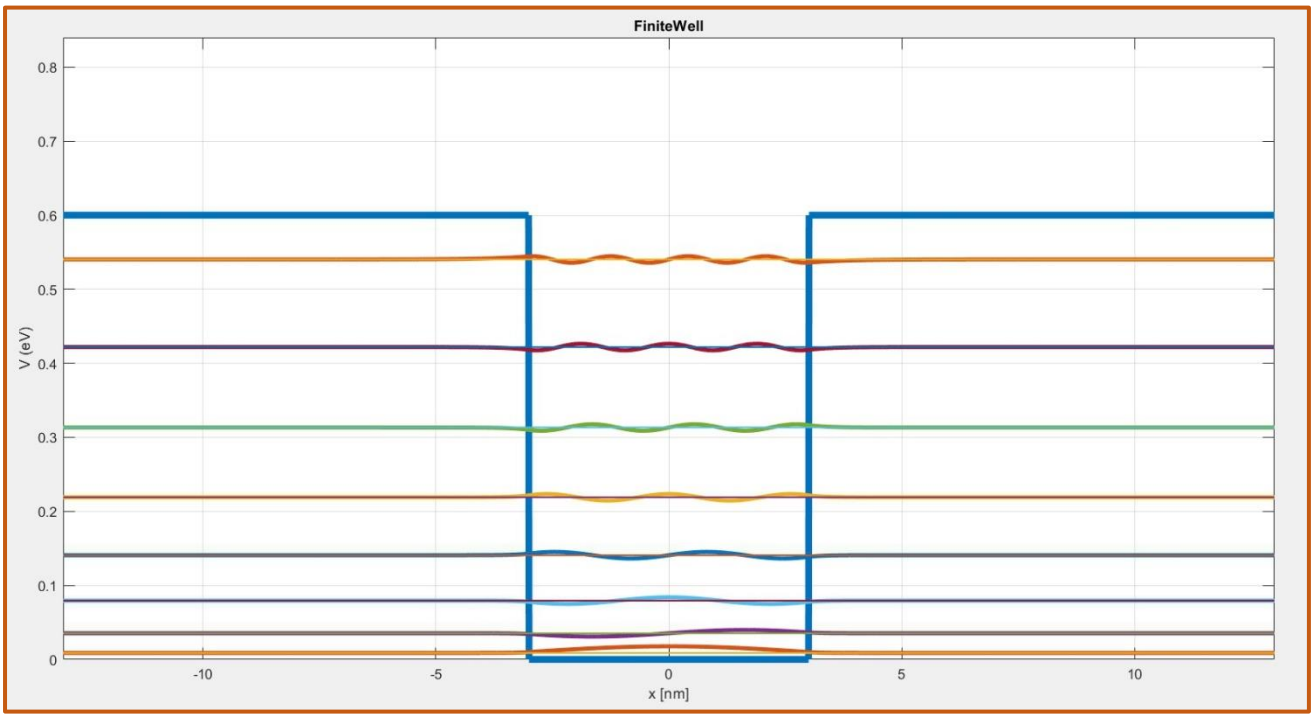
   >>1

Potential Well Width(between 1-20 (nm)): 6

Higher potential (between 0.01-1 (eV)): 0.6

Full potential well length  (between 10-50 (nm)): 26

Lowest difference (between 0.001-0.1 (nm)): 0.01
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 2: (Finite Well)

```
Command Window
================================================================================
                                Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

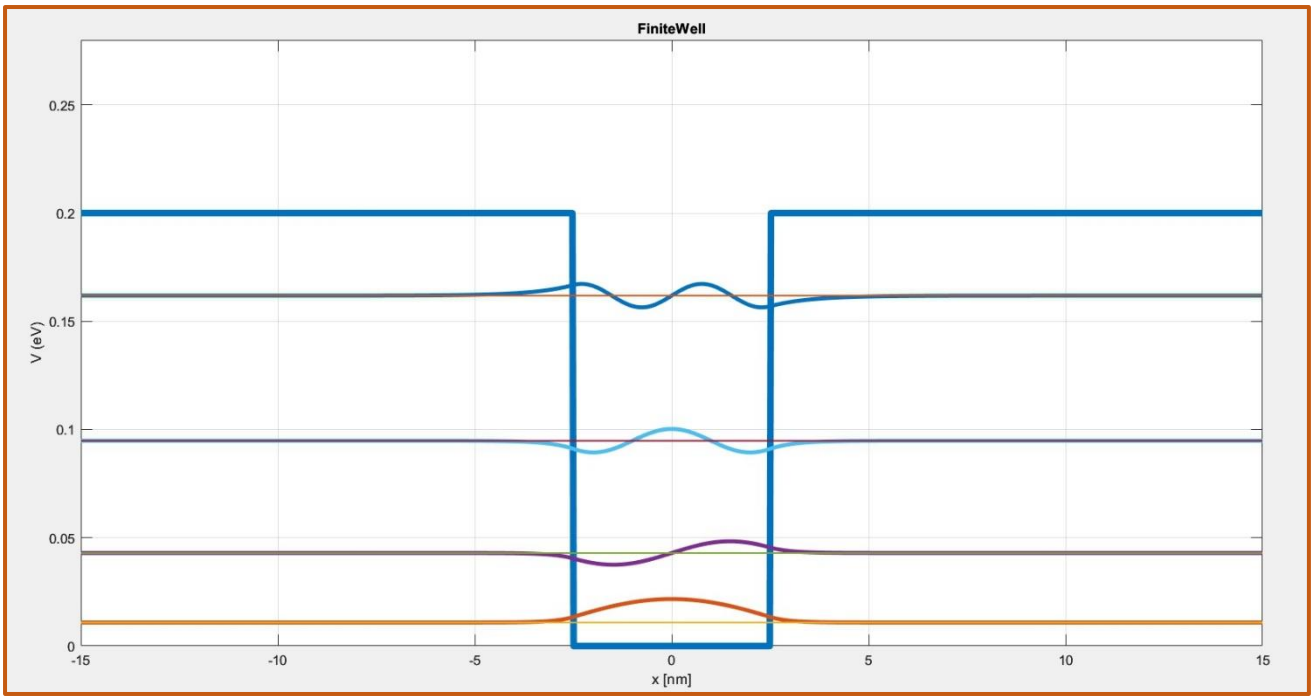
   >>1

Potential Well Width(between 1-20 (nm)): 5

Higher potential (between 0.01-1 (eV)): 0.2

Full potential well length  (between 10-50 (nm)): 30

Lowest difference (between 0.001-0.1 (nm)): 0.03
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 3: (Finite Well)

```
Command Window
================================================================================
                                Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

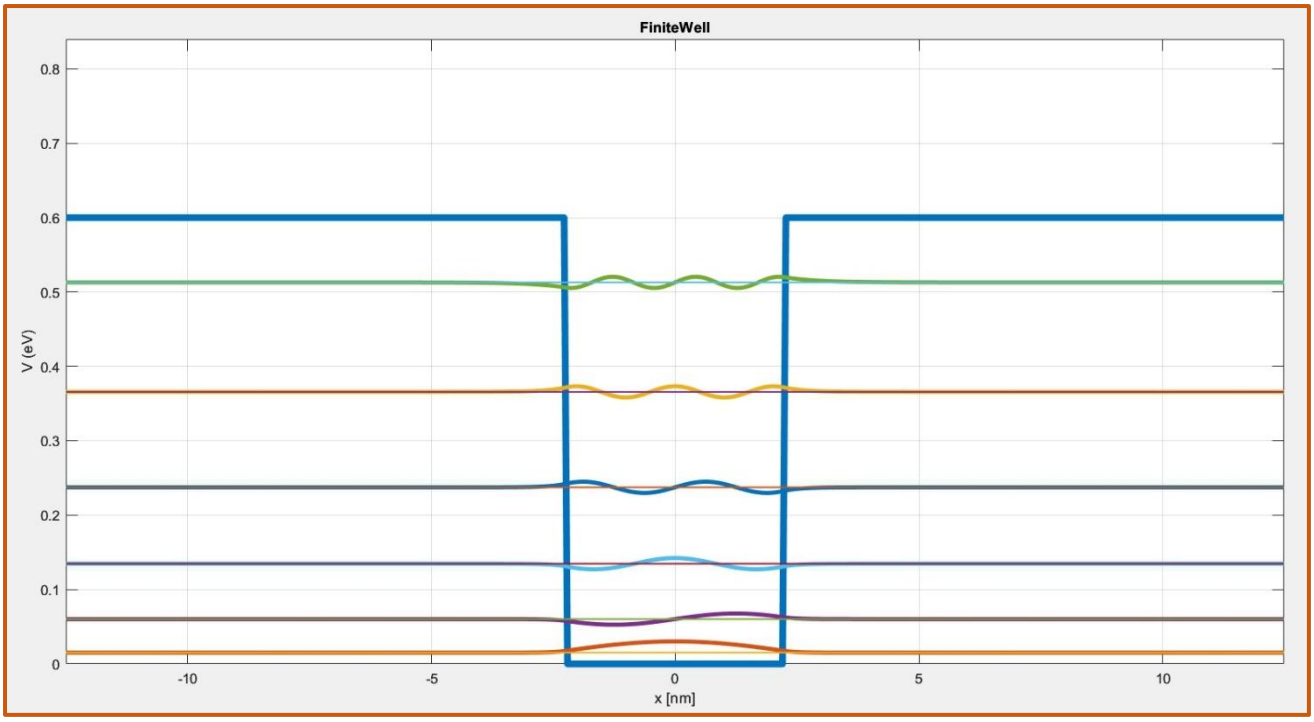
   >>1

Potential Well Width(between 1-20 (nm)): 4.5

Higher potential (between 0.01-1 (eV)): 0.6

Full potential well length  (between 10-50 (nm)): 25

Lowest difference (between 0.001-0.1 (nm)): 0.07
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 4: (Finite Well)

```
Command Window
================================================================================
                                Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

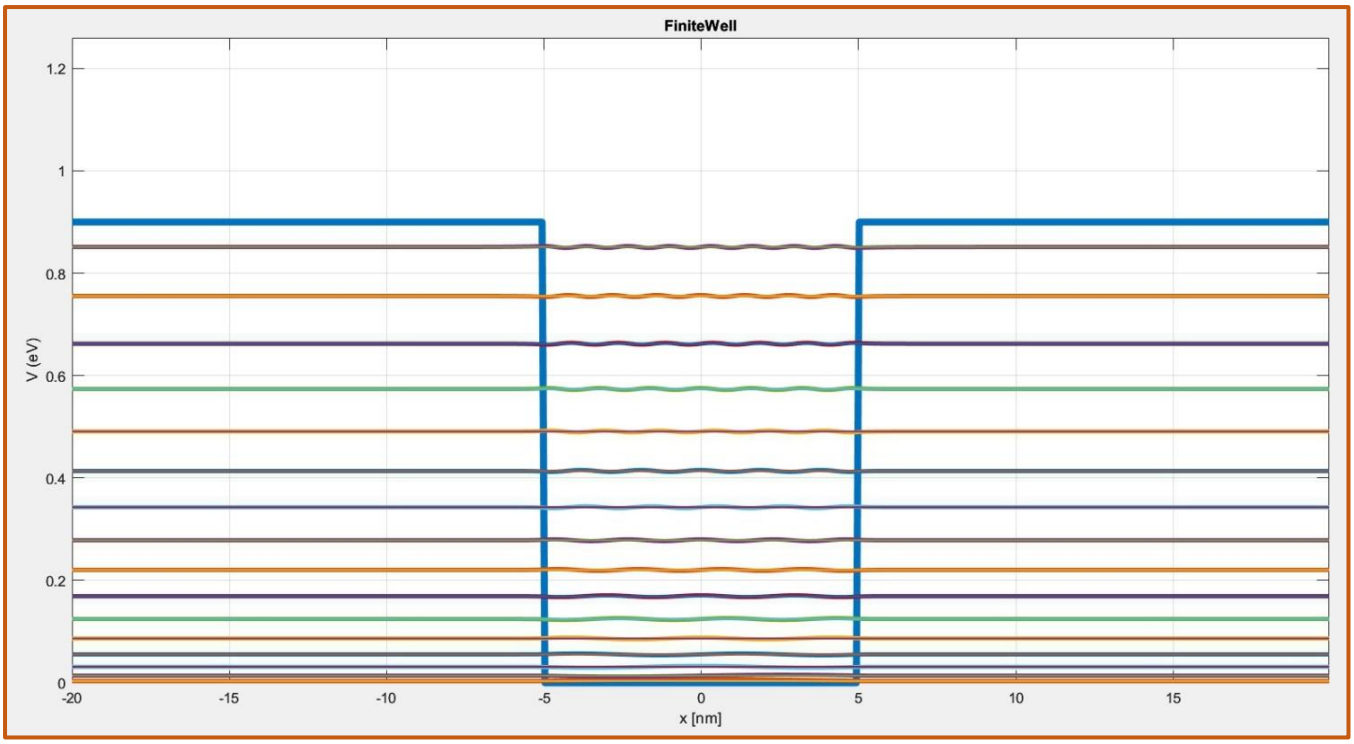
   >>1

Potential Well Width(between 1-20 (nm)): 10

Higher potential (between 0.01-1 (eV)): 0.9

Full potential well length  (between 10-50 (nm)): 40

Lowest difference (between 0.001-0.1 (nm)): 0.09
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 5 : (Infinite Well)

```
Command Window
================================================================================
                                Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

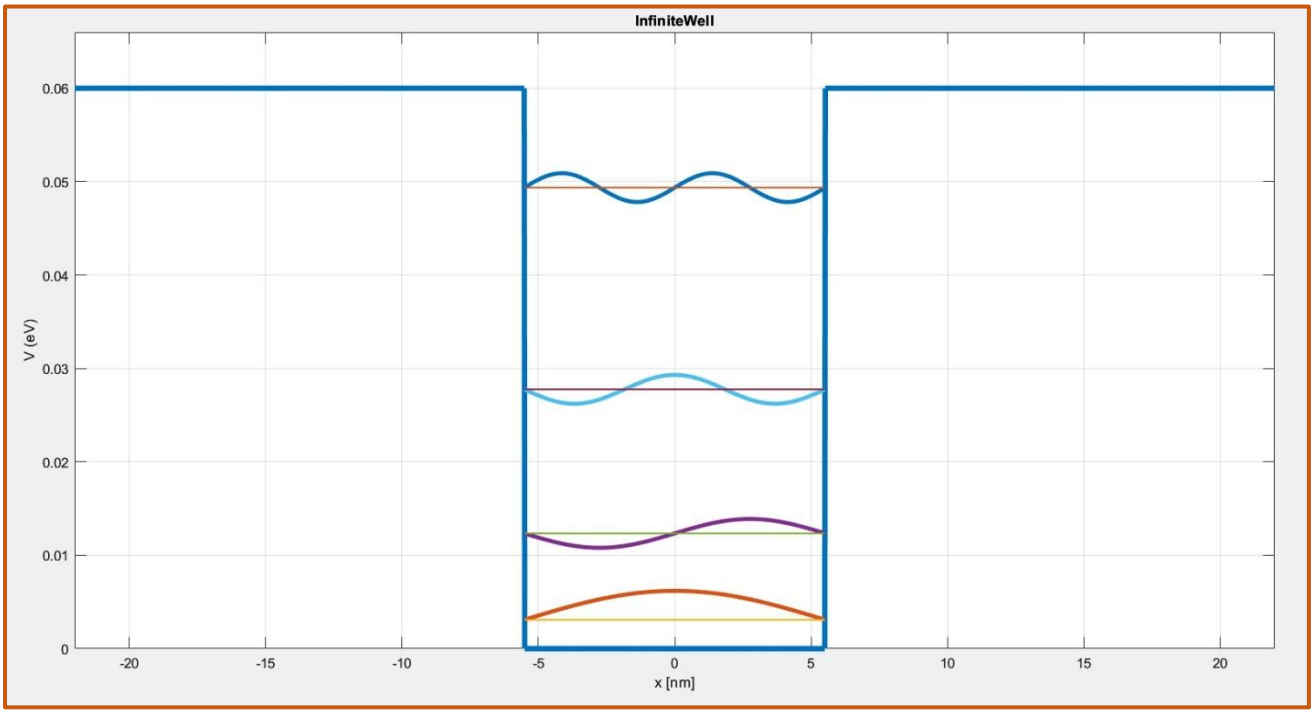
   >>2

Potential Well Width(between 1-10 (nm)): 5.5

Higher potential (between 0.01-1 (eV)): 0.06

Full potential well length  (between 10-50 (nm)): 22

Lowest difference (between 0.01-0.1 (nm)): 0.02
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 6 : (Infinite Well)

```
Command Window
New to MATLAB? See resources for Getting Started.

================================================================================
                                Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

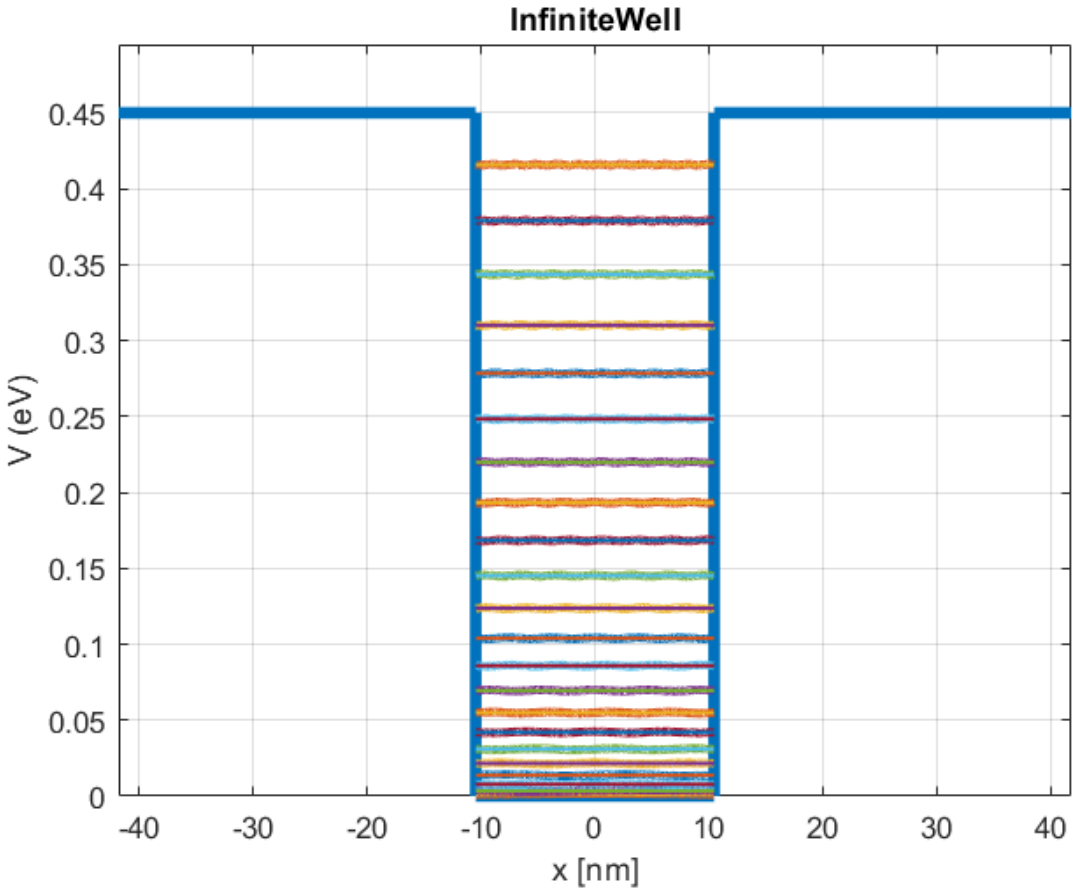
   >>2

Potential Well Width(between 1-20 (nm)): 10.45

Higher potential (between 0.01-1 (eV)): 0.45

Full potential well length  (between 10-50 (nm)): 41.8

Lowest difference (between 0.001-0.1 (nm)): 0.01
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 7 : (Infinite Well)

Command Window

New to MATLAB? See resources for Getting Started.

```
================================================================================
                                  Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

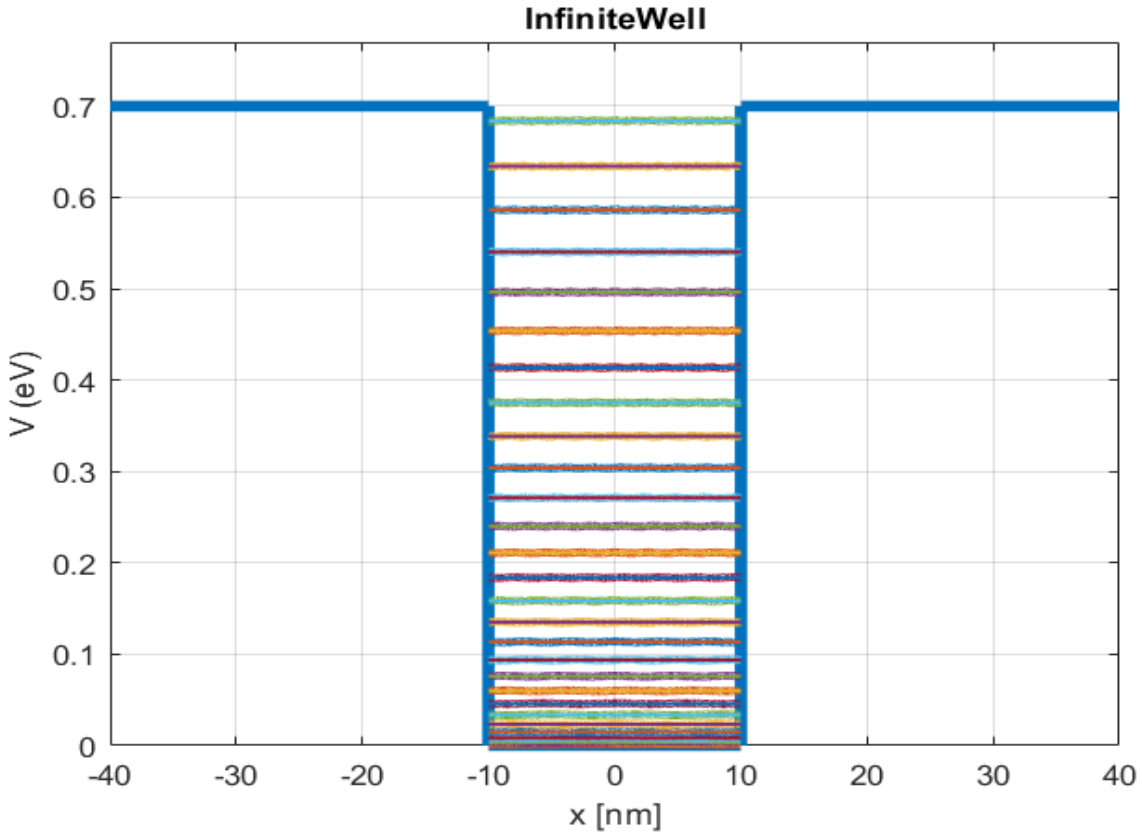
   >>2

Potential Well Width(between 1-20 (nm)): 10

Higher potential (between 0.01-1 (eV)): .7

Full potential well length  (between 10-50 (nm)): 40

Lowest difference (between 0.001-0.1 (nm)): 0.01
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 8 : (Infinite Well)

Command Window

New to MATLAB? See resources for Getting Started.

```
==========================================================================
                                Team MATLITE
==========================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

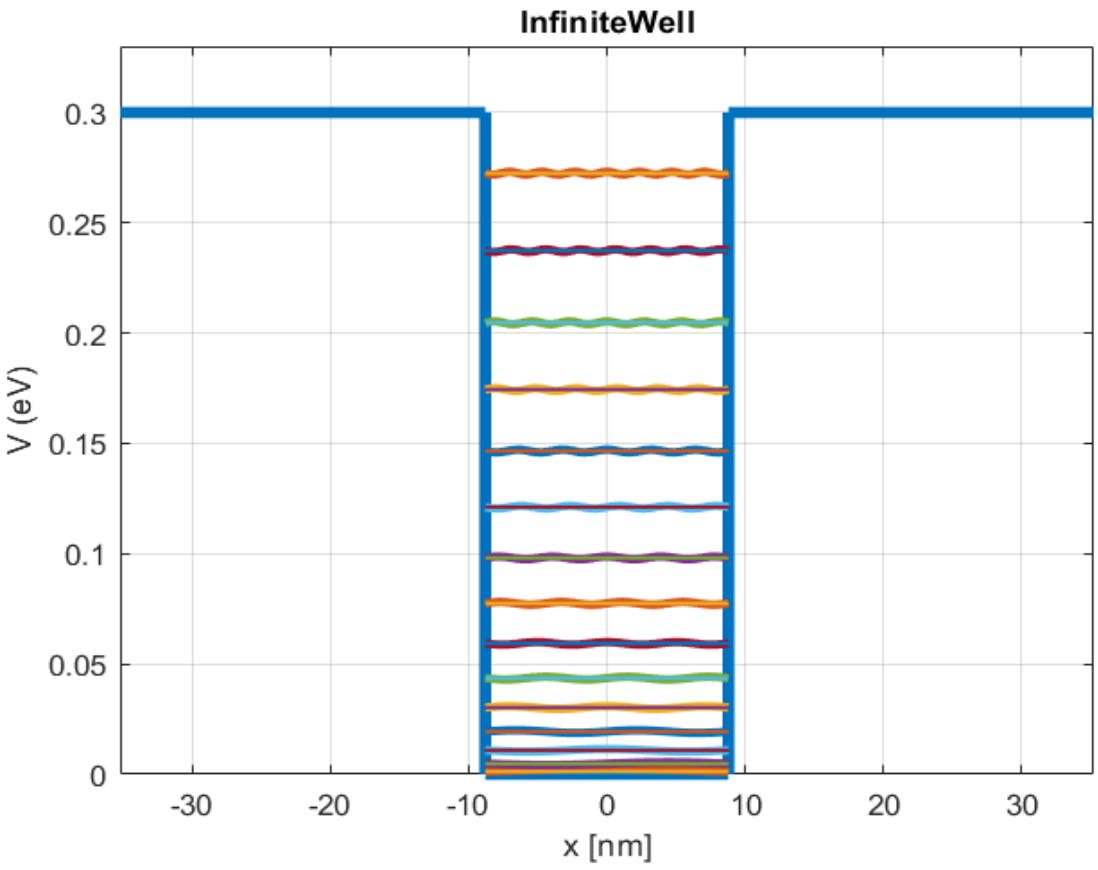
   >>2

Potential Well Width(between 1-20 (nm)): 8.8

Higher potential (between 0.01-1 (eV)): 0.3

Full potential well length  (between 10-50 (nm)): 35.2

Lowest difference (between 0.001-0.1 (nm)): 0.01
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Test conditon 9 : (Infinite Well):

Command Window

New to MATLAB? See resources for Getting Started.

```
================================================================================
                                  Team MATLITE
================================================================================
Potential Well Type:

    i.Input 1 for Finite
   ii.Input 2 for Ininite

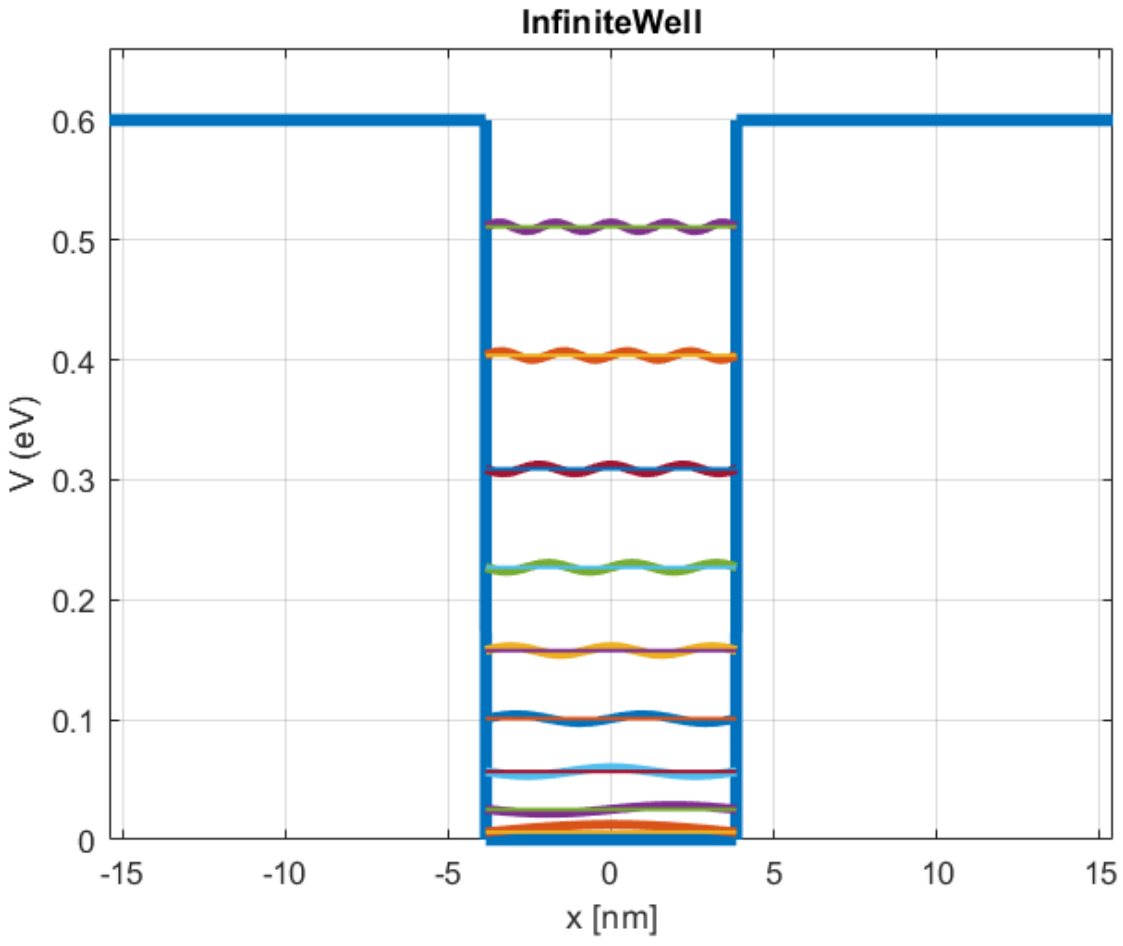
   >>2

Potential Well Width(between 1-20 (nm)): 3.85

Higher potential (between 0.01-1 (eV)): .6

Full potential well length  (between 10-50 (nm)): 15.4

Lowest difference (between 0.001-0.1 (nm)): 0.01
Team Members-
    1.Ilias Ahamd (2006068)
    2.Nahian Tasnim (2006098)
Thank you
```



Future Prospects of Improvement:

For the solution we have designed for potential wells, we could also design Graphical User Interface (GUI). Finally, using the solution we obtained here, we may try implementing it in real-life problems. For example, we can model the behavior of electrons in a semiconductor device or investigate the properties of molecules in a chemical reaction. This can help us develop new technologies or materials with specific properties.

Discussion:

Solving the time-independent Schrödinger's equation is an important problem in quantum mechanics. It allows us to determine the energy levels and wave functions of quantum systems, which are crucial in understanding the behavior of atoms, molecules, and other quantum systems. While analytical solutions to the equation are not always available, numerical methods such as the finite difference method can be used to obtain solutions.